Multi-Interval Evaluation Report

All-On Model

23 September 2026

1 Two-interval allocation

Customer type	Assigned interval	Other interval
Small	I_S	I_S
Medium	I_M	I_S
Large	I_L	I_S

Let m_1 and l_1 be the numbers of medium and large customers assigned to interval 1. The remaining $n_M - m_1$ and $n_L - l_1$ are assigned to interval 2. Thus

$$A_1 = I_M m_1 + I_L l_1 + I_S(n_S + n_M - m_1 + n_L - l_1), \tag{1}$$
$$A_2 = I_M(n_M - m_1) + I_L(n_L - l_1) + I_S(n_S + m_1 + l_1). \tag{2}$$

The model tries every integer allocation

$$m_1 \in \{0, \ldots, n_M\}, \qquad l_1 \in \{0, \ldots, n_L\}, \tag{3}$$

and chooses the allocation with the smallest maximum interval current:

$$B^*(\boldsymbol{n}) = \min_{m_1, l_1} \max(A_1, A_2). \tag{4}$$

A violation means $A_1 > T$ or $A_2 > T$. This is written as:

$$r_{\text{two}}(\boldsymbol{n}) = \begin{cases} 1, & B^*(\boldsymbol{n}) > T, \\ 0, & B^*(\boldsymbol{n}) \le T. \end{cases} \tag{5}$$

2 Three-interval allocation

Each medium or large customer is assigned to one of three intervals and draws I_S in the other two. Small customers draw I_S in all three. Let m_t, l_t be the medium and large counts assigned to interval t. Then

$$m_1 + m_2 + m_3 = n_M, \qquad l_1 + l_2 + l_3 = n_L, \tag{6}$$

where all counts are nonnegative integers. With $m_3 = n_M - m_1 - m_2$ and $l_3 = n_L - l_1 - l_2$, the currents are

$$A_1 = I_M m_1 + I_L l_1 + I_S(n_S + m_2 + m_3 + l_2 + l_3), \tag{7}$$
$$A_2 = I_M m_2 + I_L l_2 + I_S(n_S + m_1 + m_3 + l_1 + l_3), \tag{8}$$
$$A_3 = I_M m_3 + I_L l_3 + I_S(n_S + m_1 + m_2 + l_1 + l_2). \tag{9}$$

The optimal allocation minimizes the largest of the three currents:

$$B_3^*(\boldsymbol{n}) = \min_{\substack{m_1+m_2+m_3=n_M \\ l_1+l_2+l_3=n_L}} \max(A_1, A_2, A_3). \tag{10}$$

A violation occurs if any interval exceeds T, so

$$r_{\text{three}}(\boldsymbol{n}) = \begin{cases} 1, & B_3^*(\boldsymbol{n}) > T, \\ 0, & B_3^*(\boldsymbol{n}) \le T. \end{cases} \tag{11}$$

The implementation computes the exact minimum by reusing solutions for smaller remaining allocation problems. It does not use a heuristic or sampling.

3 Optimal versus random allocation

For $K \in \{1, 2, 3\}$ intervals, random allocation means each medium and large customer independently chooses one interval uniformly. We draw *one assignment per composition*, giving counts $(m_1, \ldots, m_K)$ and $(l_1, \ldots, l_K)$, where

$$\sum_{t=1}^{K} m_t = n_M, \qquad \sum_{t=1}^{K} l_t = n_L. \tag{12}$$

For this realised assignment, calculate

$$A_t^{\text{random}} = I_M m_t + I_L l_t + I_S(N - m_t - l_t). \tag{13}$$

The output is a binary violation outcome, not an average over assignments:

$$v_{\text{random},K}(\boldsymbol{n}) = \begin{cases} 1, & \max_t A_t^{\text{random}} > T, \\ 0, & \max_t A_t^{\text{random}} \le T. \end{cases} \tag{14}$$

4 Evaluation

The calculation uses

$$(I_S, I_M, I_L) = (20, 40, 100)\ \text{A}, \qquad N = 40, \qquad T = 1000\ \text{A}. \tag{15}$$

Outcome	Compositions
Already safe with one shared interval	21
Made safe by two-interval allocation	44
Still violating after optimal allocation	796

Intervals	Optimal: points reversed	Random: points reversed
1	0	0
2	44	24
3	112	48

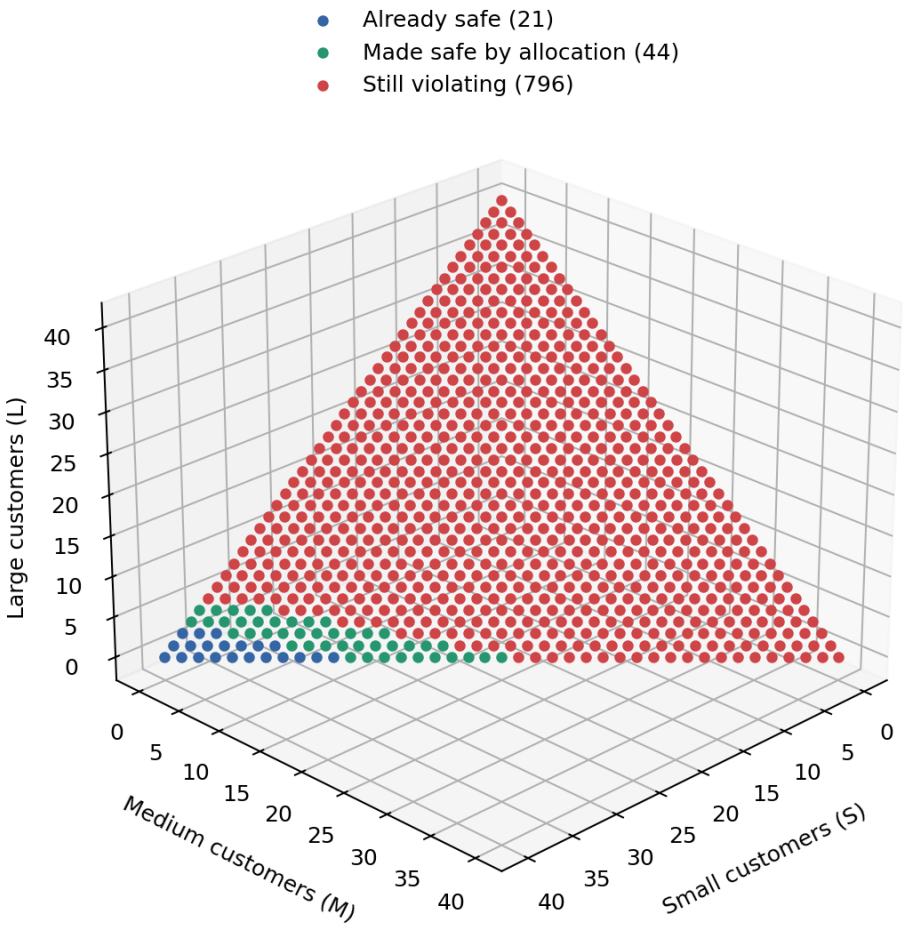


Figure 1: All-On two-interval allocation results. Colours show the outcome for each composition, not its trinomial probability.

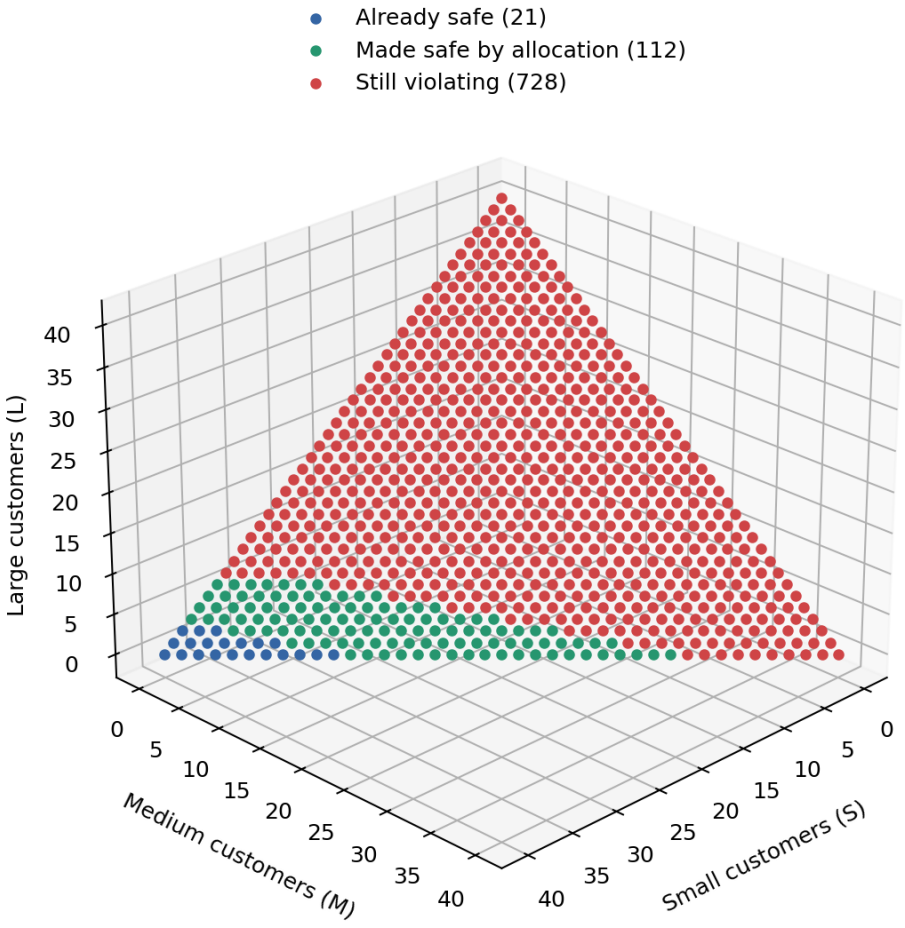


Figure 2: All-On three-interval optimal allocation results.